

Data

Sample structure

Data file: hw3_data_s9618.csv
Sample size: $n = 100$ paired daily returns
Variables: asset1_return, asset2_return
Bootstrap unit: full paired row

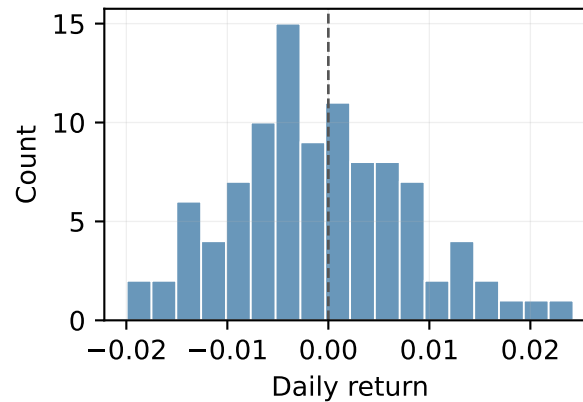
Daily return summary

Asset	Mean	SD	Min	Median	Max
Asset 1	-0.0971%	0.8804%	-1.9967%	-0.1920%	2.4270%
Asset 2	0.0179%	1.3880%	-3.6253%	0.0701%	3.5812%

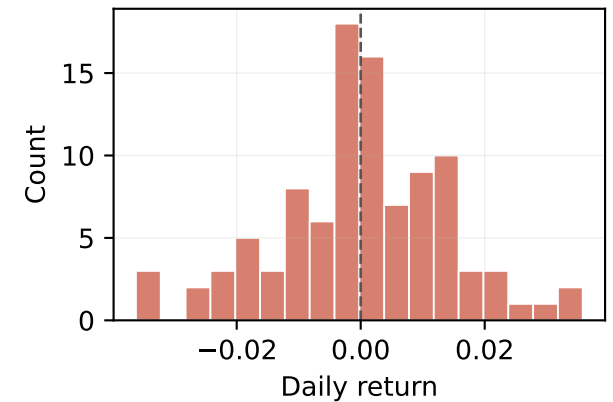
Variance and dependence

Quantity	Value
Asset 1 variance	0.00007751
Asset 2 variance	0.00019265
Covariance	-0.00000437
Correlation	-0.0358

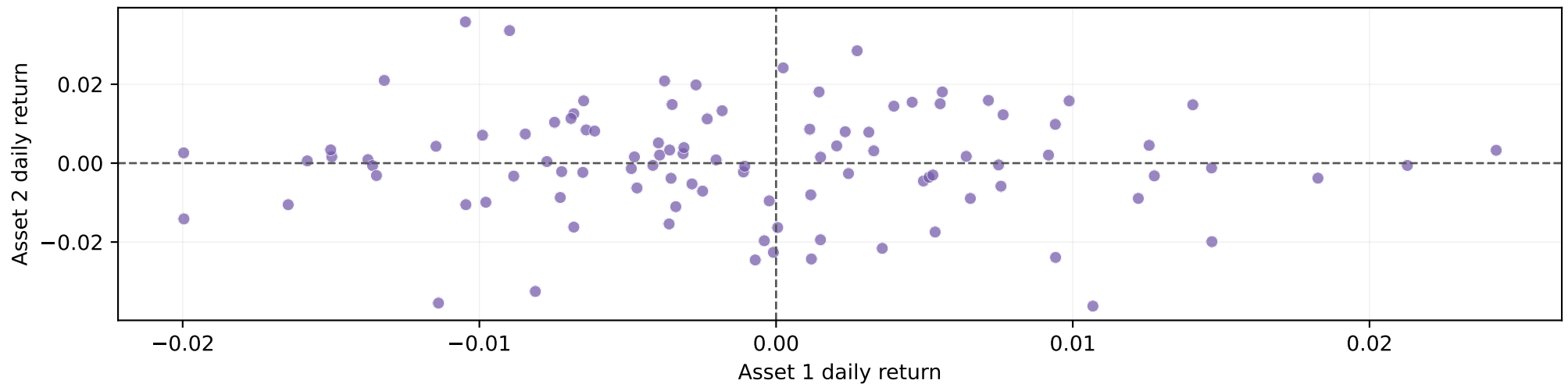
Asset 1 returns



Asset 2 returns



Paired return cloud (correlation = -0.036)



Model, Method and Results

Model formula

Minimum-variance allocation:

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

Portfolio return:

$$R_p(\alpha) = \alpha X + (1 - \alpha)Y$$

Plug-in estimate uses sample variance and covariance.

Bootstrap method

Nonparametric bootstrap from the EDF:

1. Generate U from Uniform(0,1)
2. Convert to row index: floor(n*U)
3. Keep paired returns together
4. Recompute alpha* for each replicate
5. Use 2.5% and 97.5% alpha* percentiles

Libraries: pandas, numpy, matplotlib

Model assumptions

Assumptions:

- the 100 daily pairs are representative of the return process
- rows are independent enough for row-level bootstrap inference
- paired row resampling preserves covariance between assets
- no parametric normal return model is required
- the sample covariance matrix is stable enough for the ratio estimate

Final estimate and interval

Quantity	Value
alpha_hat	0.706418
Allocation to Asset 2	0.293582
95% CI lower	0.612544
95% CI upper	0.791623
CI width	0.179079
Bootstrap B	50000

Percentile CI calculation

Bootstrap alpha* quantile	Value
2.5%	0.612544
50.0%	0.706524
97.5%	0.791623

Interpretation

The fitted minimum-variance portfolio puts 70.6% in Asset 1 and 29.4% in Asset 2.

The bootstrap interval [0.613, 0.792] is fairly tight relative to the 0 to 1 allocation scale, so the data consistently favor a larger weight on Asset 1.

The interval is still reported on alpha directly, allowing the grader to evaluate coverage and interval width.

Visualization

